

Recurrence Relations

①

Recurrence Relation for sequence $\{a_n\}$ is an equation that expresses a_n in terms of 1 or more of previous terms of the sequence namely $a_0, a_1, \dots, a_{n-1}$ for all integers n with $n \geq n_0$ where n_0 is a non-negative integer.

Eg let $\{a_n\}$ be sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ & suppose that $a_0 = 3$ & $a_1 = 5$. What are a_2 & a_3 ?

soln $a_n = a_{n-1} - a_{n-2}$

for a_2

$$\begin{aligned} a_2 &= a_{2-1} - a_{2-2} \\ &= a_1 - a_0 \\ &= 5 - 3 \end{aligned}$$

$$\boxed{a_2 = 2}$$

for a_3

$$\begin{aligned} a_3 &= a_{3-1} - a_{3-2} \\ &= a_2 - a_1 \\ &= 2 - 5 \\ &= -3 \end{aligned}$$

Solution :- A sequence is called a solution of recurrence relation if its terms satisfy the recurrence relation.

②

Equation from relation is called as Difference equation

⇒ Order of Recurrence Relation

Difference b/w Greatest suffix & least suffix

Eg = Let $a_n = a_{n-1} + a_{n-2}$ then order $n - (n-2) = 2$

⇒ Linear Recurrence Relation → Homogenous
→ Non Homogenous

A recurrence relation having only the first power of sequence $\langle a_n \rangle$ & not having any product terms is called as linear.

{ No square or cube or any product }

Eg $5a_n - 3a_{n-1} + a_{n-2} = 0$

⇒ Linear Recurrence Relation with Constant Coefficient
A recurrence relation of the form

$$C_0 a_n + C_1 a_{n-1} + \dots + C_k a_{n-k} = f(n) \quad \text{--- (1)}$$

⇓

It is known as k^{th} order recurrence relation provided $C_0 \neq 0$ & $C_k \neq 0$

$C_0 a_n + C_1 a_{n-1} = f(n)$ → first order recurrence relation

The solution of equation (1) is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

where $a_n^{(h)}$ = Homogeneous solution
 $a_n^{(p)}$ = Particular solution

a) Homogeneous recurrence relation

If $f(n) = 0$ then eqⁿ 1 is called Homogeneous recurrence relation

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = 0$$

b) Non Homogeneous Recurrence relation

If $f(n) \neq 0$ then eqⁿ 1 is called Non Homogeneous

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \neq 0 \text{ or } f(n)$$

Homogeneous Recurrence Relation with Constant Coefficients.

Suppose second order Homogeneous ^{linear recurrence} relation is

$$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} = 0$$

The characteristic Eqⁿ (Auxiliary Eqⁿ) is

$$C_0 m^2 + C_1 m + C_2 = 0$$

Case I If the roots of A.E are real & unequal
 $m_1 \neq m_2$

General solⁿ is

$$a_n = C_1 m_1^n + C_2 m_2^n$$

Case II If the roots of A.E are Real & Equal
Say $m_1 = m_2 = m$

General solⁿ is

$$a_r = (C_1 + r C_2) m^r$$

Case III If the roots of A.E are in complex no
 $m = \alpha \pm i\beta$

General solⁿ is

$$a_r = (C_1 \cos r\theta + C_2 \sin r\theta) R^r$$

where

$$R = \sqrt{\alpha^2 + \beta^2}$$

$$\theta = \tan^{-1} \left(\frac{\beta}{\alpha} \right)$$

Q1 Solve the Recurrence Relation

$$a_n + 5a_{n-1} + 6a_{n-2} = 0$$

Solⁿ This relation is of second order $a_r + 5a_{r-1} + 6a_{r-2}$

$$a_r + 5a_{r-1} + 6a_{r-2} \quad \text{--- (1)}$$

Characteristic Eqⁿ is

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m = -2, -3$$

→ Both are distinct

General solⁿ is in the form of $a_r = C_1 m_1^r + C_2 m_2^r$

$$a_r = C_1 (-2)^r + C_2 (-3)^r \quad \text{Ans}$$

Q2 Solve the Recurrence relation

$$a_r - 7a_{r-1} + 10a_{r-2} = 0$$

given that $a_0 = 0$ $a_1 = 3$

Initial Condⁿ

$$a_r - 7a_{r-1} + 10a_{r-2} = 0 \quad \text{--- (1)}$$

Given $a_0 = 0$ $a_1 = 3$

This is 2nd order recurrence relation

Characteristic equation is

$$m^2 - 7m + 10 = 0$$

$$(m-2)(m-5) = 0$$

$$m = 2, 5$$

Both are distinct

General solⁿ is

$$a_r = c_1 2^r + c_2 5^r \quad - (2)$$

Putting in eqⁿ 2 $a_0 = 0$ i.e. ~~is~~ $a_r = 0$ & $r = 0$

$$0 = c_1 2^0 + c_2 5^0$$

$$0 = c_1 + c_2 \quad - (3)$$

Again putting in eqⁿ 2 $a_1 = 3$ i.e. $a_r = 3$ & $r = 1$

$$c_1 (2)' + c_2 (5)' = 3$$

$$2c_1 + 5c_2 = 3 \quad - (4)$$

Solve (3) & (4)

$$c_1 + c_2 = 0$$

$$2c_1 + 5c_2 = 3$$

~~Putting~~ $5c_1 + 5c_2 = 0$

$$2c_1 + 5c_2 = 3$$

$$3c_1 = 3$$

$$3c_1 = -3$$

$$c_1 = -1$$

$$2(-1) + 5c_2 = 3$$

$$-2 + 5c_2 = 3$$

$$5c_2 = 5$$

$$c_2 = 5/5$$

$$c_2 = 1$$

Put c_1 & c_2 in eqⁿ 2
The required general solⁿ

$$a_r = 5^r - 2^r$$

(5)

Q3 Solve the recurrence relation

$$a_r - 3a_{r-1} + 3a_{r-2} - a_{r-3} = 0 \text{ given that } a_0 = 1$$

$$a_1 = -2 \text{ \& } a_2 = -1$$

Solⁿ Given

$$a_r - 3a_{r-1} + 3a_{r-2} - a_{r-3} = 0 \quad \text{--- (1)}$$

This is 3rd order recurrence relation

$$\text{Given that } a_0 = 1, a_1 = -2 \text{ \& } a_2 = -1$$

The characteristic eqⁿ is

$$m^3 - 3m^2 + 3m - 1 = 0$$

$$\underbrace{\hspace{10em}}_{(m-1)^3} = m^3 - 3m^2 + 3m - 1$$

$$(m-1)^3 = 0$$

$$m = 1, 1, 1$$

General solⁿ is: This is case II $m_1 = m_2 = m_3$

$$a_r = (c_1 + c_2 r + c_3 r^2)(1)^r$$

$$a_r = c_1 + c_2 r + c_3 r^2 \quad \text{--- (2)}$$

1st condition is $a_0 = 1$ means $r = 0, a_r = 1$ Putting in eqⁿ 2

$$1 = c_1 + 0 + 0$$

$$\boxed{c_1 = 1}$$

2nd condⁿ is $a_1 = -2$ where $a = 1$ & $a_r = -2$

Putting in eqⁿ 2

$$a_r = c_1 + c_2 r + c_3 r^2$$

$$-2 = c_1 + c_2(1) + c_3(1)^2$$

$$-2 = c_1 + c_2 + c_3 \quad \text{--- (3)}$$

Since $c_1 = 1$

$$-2 = 1 + c_2 + c_3$$

$$-3 = c_2 + c_3$$

3rd condⁿ is $a_2 = -1$ where $a = 2$ & $a_r = -1$

Putting in eqⁿ 2

$$a_r = c_1 + c_2 r + c_3 r^2$$

$$-1 = c_1 + c_2(2) + c_3(2)^2$$

$$-1 = c_1 + 2c_2 + 4c_3 \quad \text{--- (4)}$$

Solving eqⁿ (3) & (4)

$$c_1 + c_2 + c_3 = -2$$

$$c_1 + 2c_2 + 4c_3 = -1$$

$$-c_2 - 3c_3 = -1$$

$$c_2 + 3c_3 = 1$$

$$\boxed{c_2 = -11/2 \quad \& \quad c_3 = 5/2}$$

Putting value of c_1, c_2, c_3 in eqⁿ 2

$$a_r = 1 - \frac{11}{2}r + \frac{5}{2}r^2$$

Q4 Solve the recurrence relation

$$a_n + 2a_{n-1} + 2a_{n-2} = 0 \quad \text{given } a_0 = 0, a_1 = -1$$

Solⁿ $a_r + 2a_{r-1} + 2a_{r-2} = 0$

This is 2nd order recurrence relation

$$m^2 + 2m + 2 = 0 \quad \{ \text{characteristic eqⁿ} \}$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-2 \pm \sqrt{(2)^2 - 4 \times 1 \times 2}}{2 \times (1)}$$

$$= \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$m = -1 \pm i$$

This is in case 3 i.e. $m = \alpha \pm i\beta$

$$m = -1 \pm i$$

$$\alpha = -1 \quad \beta = 1$$

$$R = \sqrt{\alpha^2 + \beta^2}$$

$$= \sqrt{(-1)^2 + (1)^2}$$

$$R = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$$

$$= \tan^{-1}\left(\frac{-1}{1}\right)$$

$$\theta = \tan^{-1}(-1)$$

$\tan^{-1}$ is $\pi/4$ so

$$\theta = \pi - \frac{\pi}{4}$$

$$\theta = \frac{3\pi}{4}$$

$\therefore$ General solⁿ is

$$a_r = (c_1 \cos r\theta + c_2 \sin r\theta) R^r$$

$$= \left[c_1 \cos\left(\frac{3\pi}{4} r\right) + c_2 \sin\left(\frac{3\pi}{4} r\right) \right] (\sqrt{2})^r$$

Putting in eqⁿ (2) $a_0 = 0$ i.e. $r=0$ $a_r = 0$

$$0 = [c_1 \cos 0 + c_2 \sin 0] (\sqrt{2})^0$$

$$0 = [c_1 + 0] (1)$$

$$\boxed{0 = c_1}$$

Putting in eqⁿ (2) $a_1 = -1$ i.e. $r=1$ $a_r = -1$

$$-1 = \left[c_1 \cos\left(\frac{3\pi}{4}\right) + c_2 \sin\left(\frac{3\pi}{4}\right) \right] (\sqrt{2})^1$$

Since $c_1 = 0$

$$-1 = \left[c_2 \sin \frac{3\pi}{4} \right] \sqrt{2}$$

$$-1 = \left[0 + c_2 \frac{1}{\sqrt{2}} \right] (\sqrt{2})^1$$

$$\boxed{-1 = c_2}$$

Putting c_1 & c_2 in eqⁿ (2)

$$a_r = \left[0 + (-1) \sin\left(\frac{3\pi}{4} r\right) \right] (\sqrt{2})^r$$

$$\boxed{a_r = -\sin\left(\frac{3\pi}{4} r\right) (\sqrt{2})^r} \text{ Ans}$$